

## ILE SURPRISE, New Caledonia

WMO: 915700

Lat: 18.480S

Lon: 163.086E

Elev: 4

StdP: 101.28

Time Zone: 11.00 (E11)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 8             | 19.8       | 20.4 | 12.9                          | 9.3 | 21.8 | 13.8 | 9.8 | 22.1 | 12.1                  | 22.6 | 11.2 | 23.2 | 3.7                   | 150  | 0.476 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 5.5                    | 33.2            | 26.8 | 32.5 | 26.4 | 31.8 | 26.1 | 28.2                | 31.6 | 27.7 | 30.8 | 27.2 | 30.2 | 2.4                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.2                            | 23.0 | 30.2 | 26.8 | 22.4 | 29.6 | 26.4 | 21.9 | 29.2 | 90.6          | 31.9 | 88.2 | 30.9 | 86.3 | 30.2 | 33.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 10.8              | 10.0 | 9.3 | 17.5                       | 33.8 | 1.7                | 1.4 | 16.3                                               | 34.8 | 15.3       | 35.6 | 14.3       | 36.3 | 13.1       | 37.3 |
|                   |      |     | 14.9                       | 28.8 | 1.1                | 0.6 | 14.1                                               | 29.2 | 13.4       | 29.5 | 12.8       | 29.8 | 12.0       | 30.2 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 26.3        | 28.2  | 28.7  | 28.2  | 27.4  | 26.1  | 24.7  | 23.8  | 23.7  | 24.6  | 25.9  | 26.9  | 27.9  | (6)  |
|                                                                        | DBStd       | 2.10        | 1.12  | 1.18  | 1.09  | 1.30  | 1.19  | 1.21  | 1.19  | 1.31  | 1.19  | 1.22  | 1.02  | 1.11  | (7)  |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |
|                                                                        | HDD18.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (9)  |
|                                                                        | CDD10.0     | 5962        | 565   | 524   | 565   | 523   | 498   | 441   | 427   | 424   | 437   | 494   | 507   | 556   | (10) |
|                                                                        | CDD18.3     | 2921        | 307   | 291   | 307   | 273   | 240   | 191   | 169   | 166   | 187   | 236   | 257   | 298   | (11) |
|                                                                        | CDH23.3     | 24042       | 3363  | 3335  | 3314  | 2638  | 1743  | 884   | 494   | 527   | 872   | 1615  | 2192  | 3066  | (12) |
|                                                                        | CDH26.7     | 6530        | 1134  | 1228  | 1013  | 646   | 251   | 69    | 24    | 42    | 123   | 382   | 592   | 1026  | (13) |
| (14) Wind                                                              | WSAvg       | 4.9         | 4.3   | 4.3   | 5.1   | 5.5   | 4.6   | 5.0   | 5.0   | 4.7   | 4.8   | 5.6   | 5.2   | 4.7   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 1072        | 129   | 138   | 181   | 130   | 87    | 75    | 51    | 50    | 37    | 40    | 66    | 88    | (15) |
|                                                                        | PrecMax     | 1696        | 314   | 379   | 410   | 387   | 327   | 212   | 178   | 130   | 120   | 123   | 249   | 322   | (16) |
|                                                                        | PrecMin     | 539         | 9     | 12    | 25    | 10    | 9     | 8     | 4     | 9     | 7     | 4     | 9     | 9     | (17) |
|                                                                        | PrecStd     | 279         | 82    | 93    | 105   | 96    | 63    | 51    | 37    | 32    | 30    | 34    | 52    | 82    | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 33.7  | 34.5  | 33.6  | 34.5  | 31.6  | 29.5  | 28.3  | 29.0  | 29.7  | 31.3  | 32.2  | 33.2  | (19) |
|                                                                        |             | MCWB        | 26.8  | 27.6  | 27.3  | 28.4  | 25.8  | 24.7  | 23.2  | 23.5  | 23.1  | 24.1  | 25.5  | 25.8  | (20) |
|                                                                        | 2%          | DB          | 32.7  | 33.2  | 32.2  | 31.8  | 29.9  | 28.3  | 27.3  | 27.9  | 28.9  | 30.5  | 31.2  | 32.5  | (21) |
|                                                                        |             | MCWB        | 26.3  | 27.4  | 26.8  | 26.5  | 25.2  | 24.0  | 22.4  | 22.6  | 22.8  | 24.1  | 24.9  | 25.6  | (22) |
|                                                                        | 5%          | DB          | 31.9  | 32.4  | 31.4  | 30.7  | 28.9  | 27.4  | 26.5  | 26.9  | 28.1  | 29.7  | 30.4  | 31.8  | (23) |
|                                                                        |             | MCWB        | 26.1  | 26.8  | 26.6  | 26.2  | 24.7  | 23.4  | 21.9  | 21.9  | 22.5  | 23.8  | 24.5  | 25.4  | (24) |
|                                                                        | 10%         | DB          | 31.2  | 31.5  | 30.6  | 29.8  | 28.0  | 26.6  | 25.7  | 25.9  | 27.2  | 28.9  | 29.7  | 31.1  | (25) |
|                                                                        |             | MCWB        | 26.0  | 26.5  | 26.4  | 25.8  | 24.0  | 22.9  | 21.6  | 21.3  | 22.1  | 23.3  | 24.1  | 25.1  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 28.3  | 28.8  | 28.5  | 29.4  | 27.1  | 26.1  | 25.1  | 24.9  | 24.9  | 26.0  | 27.2  | 27.6  | (27) |
|                                                                        |             | MCDB        | 31.7  | 32.7  | 32.0  | 33.1  | 29.8  | 28.2  | 26.7  | 27.4  | 28.0  | 29.5  | 30.4  | 31.4  | (28) |
|                                                                        | 2%          | WB          | 27.7  | 28.2  | 27.8  | 27.5  | 26.2  | 25.2  | 24.2  | 24.1  | 24.1  | 25.2  | 26.2  | 26.9  | (29) |
|                                                                        |             | MCDB        | 30.9  | 31.8  | 30.7  | 30.5  | 28.5  | 26.8  | 25.7  | 26.2  | 27.3  | 28.7  | 29.4  | 30.5  | (30) |
|                                                                        | 5%          | WB          | 27.1  | 27.7  | 27.4  | 26.9  | 25.6  | 24.6  | 23.4  | 23.3  | 23.5  | 24.6  | 25.5  | 26.3  | (31) |
|                                                                        |             | MCDB        | 30.3  | 31.0  | 30.0  | 29.6  | 27.7  | 26.3  | 25.2  | 25.4  | 26.5  | 28.1  | 28.7  | 29.9  | (32) |
|                                                                        | 10%         | WB          | 26.6  | 27.2  | 27.0  | 26.3  | 25.0  | 23.9  | 22.6  | 22.6  | 22.9  | 24.1  | 25.0  | 25.9  | (33) |
|                                                                        |             | MCDB        | 29.6  | 30.2  | 29.5  | 28.6  | 27.0  | 25.9  | 24.8  | 24.9  | 25.7  | 27.4  | 28.0  | 29.3  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.6   | 5.5   | 4.8   | 4.6   | 4.1   | 3.9   | 4.1   | 4.7   | 5.5   | 5.9   | 5.9   | 6.1   | (35) |
|                                                                        |             | MCDBR       | 7.0   | 6.7   | 6.0   | 6.1   | 5.3   | 4.7   | 5.0   | 5.7   | 6.6   | 6.8   | 6.9   | 7.1   | (36) |
|                                                                        | 5% WB       | MCWBR       | 2.8   | 2.7   | 2.6   | 3.0   | 2.8   | 2.6   | 2.7   | 3.0   | 3.1   | 2.8   | 2.8   | 2.8   | (37) |
|                                                                        |             | MCWBR       | 5.8   | 5.9   | 5.1   | 5.3   | 4.6   | 4.1   | 4.2   | 4.7   | 5.6   | 5.9   | 6.1   | 6.1   | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.391 | 0.390 | 0.389 | 0.382 | 0.373 | 0.373 | 0.363 | 0.361 | 0.369 | 0.385 | 0.389 | 0.381 | (40) |
|                                                                        |             | taud        | 2.595 | 2.605 | 2.616 | 2.618 | 2.607 | 2.600 | 2.596 | 2.601 | 2.555 | 2.538 | 2.551 | 2.604 | (41) |
|                                                                        | Ebn at Noon |             | 954   | 944   | 922   | 892   | 864   | 845   | 865   | 900   | 927   | 938   | 949   | 964   | (42) |
|                                                                        |             | Edh at Noon | 105   | 103   | 98    | 92    | 87    | 85    | 87    | 92    | 103   | 109   | 109   | 104   | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 6.71  | 6.49  | 5.53  | 4.86  | 4.10  | 3.77  | 4.00  | 4.80  | 5.85  | 6.47  | 6.66  | 7.00  | (44) |
|                                                                        | RadStd      |             | 0.61  | 0.57  | 0.49  | 0.48  | 0.31  | 0.22  | 0.28  | 0.29  | 0.39  | 0.44  | 0.40  | 0.57  | (45) |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page